(iii) $t_2 - t_1 = 5\sqrt{2} - 3\sqrt{2} = 2\sqrt{2}$

$t_3 - t_2 = 7\sqrt{2} - 5\sqrt{2} = 2\sqrt{2}$

$t_4 - t_3 = 9\sqrt{2} - 7\sqrt{2} = 2\sqrt{2}$

Thus, the differences between consecutive terms are equal. Hence the terms of the sequence $3\sqrt{2}, 5\sqrt{2}, 7\sqrt{2}, 9\sqrt{2}, \ldots$ are in A.P.

Example 2.24 Write an A.P. whose first term is 20 and common difference is 8.

Solution First term $= a = 20$; common difference $= d = 8$

Arithmetic Progression is $a, a + d, a + 2d, a + 3d, \ldots$

In this case, we get $20, 20 + 8, 20 + 2(8), 20 + 3(8), \ldots$

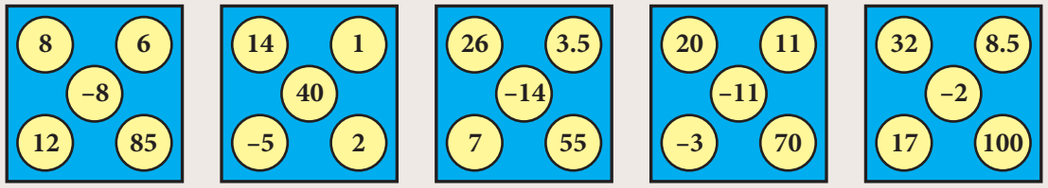
So, the required A.P. is 20, 28, 36, 44, …

Note

An Arithmetic progression having a common difference of zero is called a constant arithmetic progression.

Activity 4

There are five boxes here. You have to pick one number from each box and form five Arithmetic Progressions.



Example 2.25 Find the 15th, 24th and n^{th} term (general term) of an A.P. given by 3, 15, 27, 39,…

Solution We have, first term $= a = 3$ and common difference $= d = 15 - 3 = 12$.

We know that n^{th} term (general term) of an A.P. with first term a and common difference d is given by $t_n = a + (n-1)d$

$$t_{15} = a + (15-1)d = a + 14d = 3 + 14(12) = 171$$

(Here $a = 3$ and $d = 12$)

$$t_{24} = a + (24-1)d = a + 23d = 3 + 23(12) = 279$$

The nth (general term) term is given by $t_n = a + (n-1)d$

Thus, $t_n = 3 + (n-1)12$

$t_n = 12n - 9$

2 Numbers and Seq.indd 58 15-12-2020 02:53:04 PM